Florida State University
Department of Scientific Computing

Written Preliminary Examination

Summer 2024

Question - 4

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Contents

1 Numerical Integration[Dr. Huang]

1.1 Part (a)

An ellipsoid is defined by the equation:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} < 1$$

where a, b, and c are the semi-axis lengths along the x, y, and z axes, respectively.

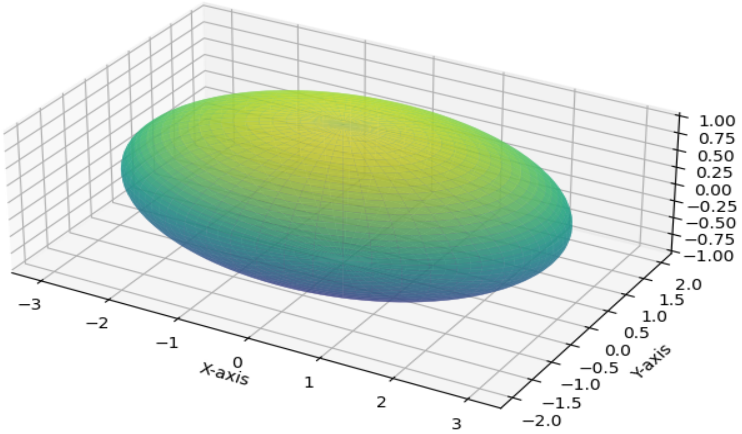

Figure 1: Ellipsoid generated using Python.

The exact volume V_{exact} of the ellipsoid is given by:

$$V_{\text{exact}} = \frac{4}{3}\pi abc$$

The domain to be considers goes from -a to a in x-direction, -b to b in y-direction and -c to c in z-direction. Hence, the volume of the bounding box of the ellipsoid is:

$$V_{\text{box}} = 8abc$$

To estimate the volume using the Monte Carlo method, I am using the following steps:

1. Generate N random points uniformly distributed within the bounding box:

$$x \sim \mathcal{U}(-a, a), \quad y \sim \mathcal{U}(-b, b), \quad z \sim \mathcal{U}(-c, c)$$

   where $\mathcal{U}$ denotes a uniform distribution.

2. Count the number of points k that lie inside the ellipsoid:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} < 1$$

3. The estimated volume V_{est} of the ellipsoid is calculated as:

$$V_{\text{est}} = V_{\text{box}} \times \frac{k}{N} = 8abc \times \frac{k}{N}$$

Given the parameters $a = 3.0$, $b = 2.0$, $c = 1.0$, the exact volume of the ellipsoid is:

$$V_{\text{exact}} = \frac{4}{3}\pi(3.0)(2.0)(1.0) = \frac{8}{3}\pi \approx 8.37758$$

For sample sizes $N \in \{1000, 10000, 100000, 1000000, 10000000\}$, the Monte Carlo method provides an estimate V_{est} and the error is:

$$\text{Error} = |V_{\text{exact}} - V_{\text{est}}|$$

N	Calculated Volume	Error	Duration (s)
1000	25.632	0.499259	0.000426526
10000	24.9408	0.191941	0.00368345
100000	25.2514	0.118619	0.0240751
1000000	25.1479	0.0151308	0.183193
10000000	25.138	0.00524277	1.86312

Table 1: Monte Carlo estimation of ellipsoid volume for different N.

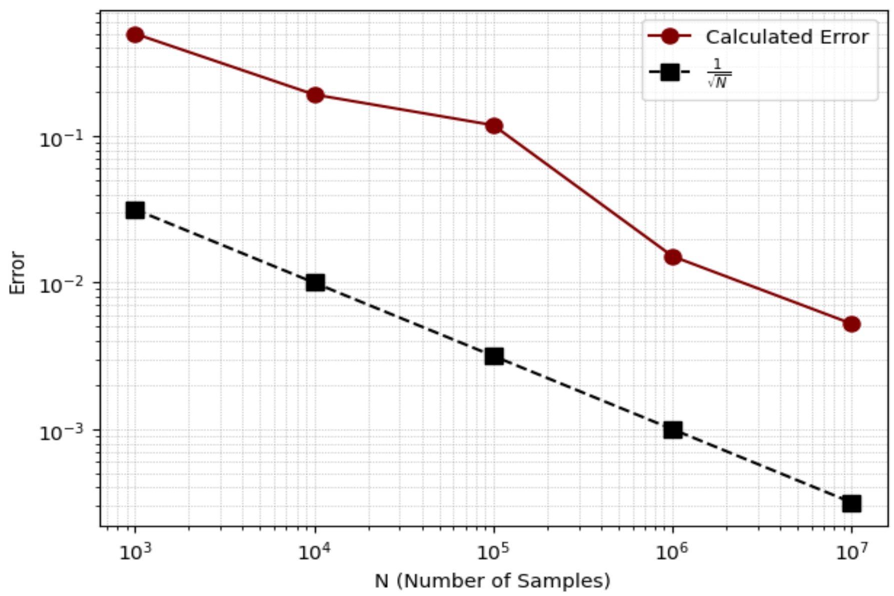

Figure 2: Error vs N plot.

From both the Table-1 and Fig-2, it is clear that error decays as $\dfrac{1}{\sqrt{N}}$.

1.2 Part (b)

The given integral is:

$$I = \int_{-5}^{0} (x^2 + 1)\, dx \tag{1}$$

To transform the integral domain from $[-5, 0]$ to $[-1, 1]$, we use the change of variable:

$$x = \frac{b-a}{2}t + \frac{b+a}{2}$$

where $a = -5$ and $b = 0$. The differential dx transforms as follows:

$$dx = \frac{b-a}{2}\,dt$$

Substituting these into the integral, we get:

$$I = \int_{-1}^{1} \left(\left(\frac{b-a}{2}t + \frac{b+a}{2} \right)^2 + 1 \right) \frac{b-a}{2}\,dt \tag{2}$$

To approximate the integral using Gauss-Legendre quadrature, we use the formula:

$$\int_a^b f(x)\,dx \approx \frac{b-a}{2} \sum_{i=1}^{n} w_i f\left(\frac{b-a}{2}x_i + \frac{b+a}{2} \right)$$

where x_i are the roots of the Legendre polynomial $P_n(x)$ and w_i are the corresponding weights.

For our integral, we have $a = -5$ and $b = 0$, thus:

$$I \approx \frac{5}{2} \sum_{i=1}^{n} w_i \left(\left(\frac{5}{2}x_i - \frac{5}{2} \right)^2 + 1 \right)$$

The exact values of x_i and w_i depend on the chosen n, the number of quadrature points.

n	Calculated Integral	Error
1	36.25	10.416699999999999
2	46.66666666666666	3.333333334154531e-05
3	46.66666666666666	3.333333334154531e-05
4	46.66666666666666	3.333333334154531e-05
5	46.66666666666668	3.333333332022903e-05

Table 2: Results of Gauss-Legendre quadrature for different values of n.

As shown in Table-1.2, the approximation of given integral is very accuarate for just 2 nodes.

Notes:

- Gauss-Legendre quadrature with n nodes is exact for polynomials of degree $2n - 1$ or less.
- For the calculation of exact integrals, I have used Integral Calculator.

1.3 Part (c)

Given the integral:

$$\int_{-\infty}^{\infty} e^{-(2x-1)^2}(x^2+1)\,dx$$

we need to transform it into a form suitable for Gauss-Hermite quadrature.

Let $u = 2x - 1$. Then:

$$x = \frac{u+1}{2}$$

The differential dx transforms as:

$$dx = \frac{du}{2}$$

Substitute $u = 2x - 1$ and $dx = \frac{du}{2}$:

$$\int_{-\infty}^{\infty} e^{-(2x-1)^2}(x^2+1)\, dx = \int_{-\infty}^{\infty} e^{-u^2}\left(\left(\frac{u+1}{2}\right)^2 + 1\right)\frac{du}{2}$$

Simplify the integrand:

$$\left(\frac{u+1}{2}\right)^2 + 1 = \frac{(u+1)^2}{4} + 1 = \frac{u^2+2u+1}{4} + 1 = \frac{u^2+2u+1+4}{4} = \frac{u^2+2u+5}{4}$$

Thus, the integral becomes:

$$\frac{1}{2}\int_{-\infty}^{\infty} e^{-u^2}\left(\frac{u^2+2u+5}{4}\right) du$$

Factor out constants from the integral:

$$\frac{1}{2}\cdot\frac{1}{4}\int_{-\infty}^{\infty} e^{-u^2}(u^2+2u+5)\, du = \frac{1}{8}\int_{-\infty}^{\infty} e^{-u^2}(u^2+2u+5)\, du$$

The integral is now in the form suitable for Gauss-Hermite quadrature:

$$I = \frac{1}{8}\int_{-\infty}^{\infty} e^{-u^2}(u^2+2u+5)\, du$$

$$\text{with,}\quad f(u) = \frac{1}{8}(u^2+2u+5)$$

n	Calculated Integral	Error
1	1.1077836568159474	0.1107783656815946
2	1.2185620224975422	2.220446049250313e-16
3	1.2185620224975422	2.220446049250313e-16
4	1.2185620224975422	2.220446049250313e-16
5	1.218562022497542	0.0
6	1.218562022497542	0.0

Table 3: Results of Gauss-Hermite quadrature for different values of n

The approximated integral results are again very accurate in this case as well and good results are obtained for just n=2.

References

- https://www.integral-calculator.com/
- https://en.wikipedia.org/wiki/Gaussian_quadrature
- ACSI Lecture Notes, Dr. Huang